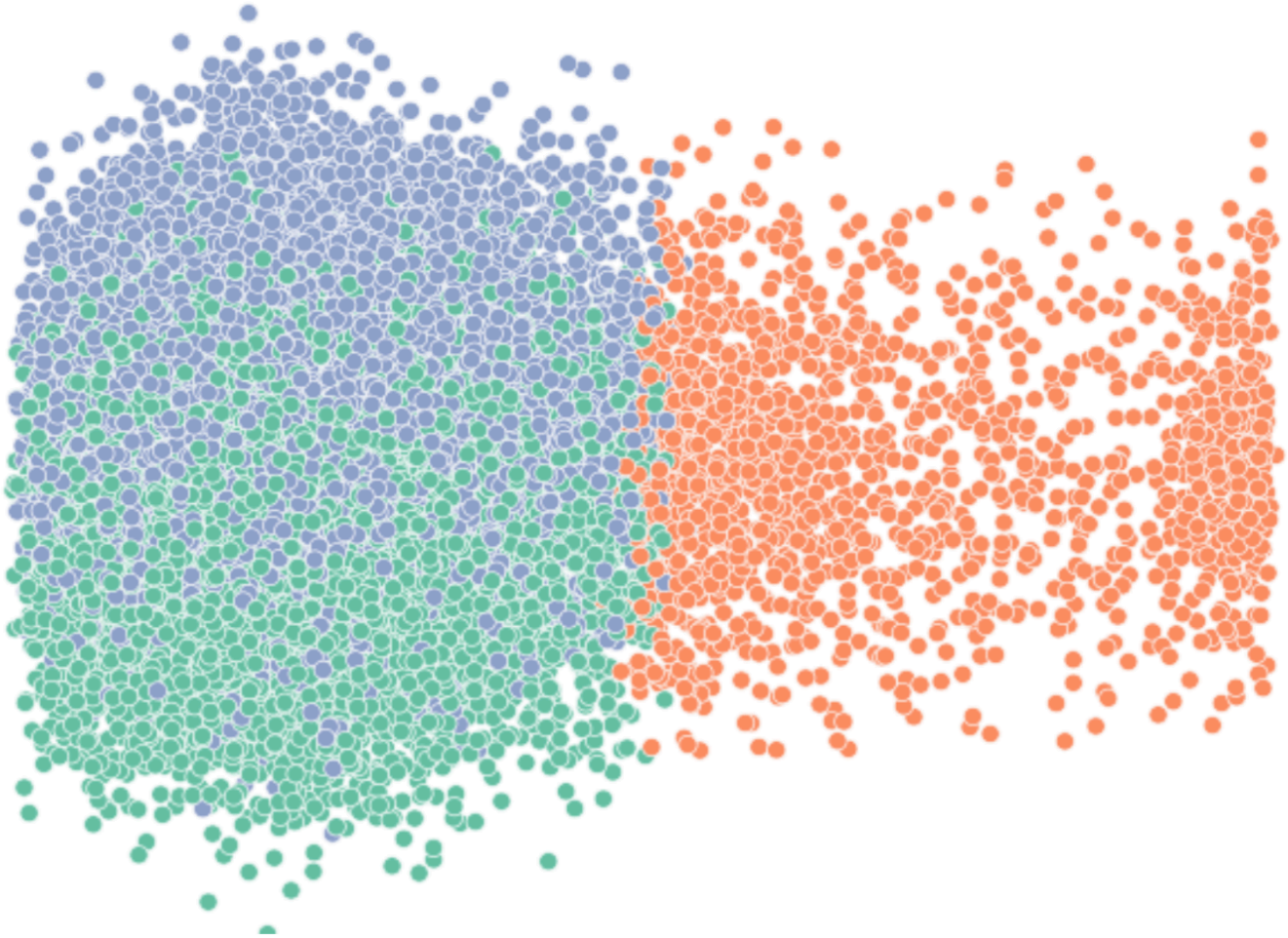
A DATA-DRIVEN ANALYSIS OF AI'S INFLUENCE ON JOB GROWTH AND DECLINE (2024–2030)

Final Project Report

Introduction to Data Science - COMP 5500



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1. Introduction

In this project, I explored how artificial intelligence might influence the job market between 2024 and 2030. I chose this topic because I am entering the IT and data science field myself, and I wanted to understand whether AI will create more opportunities or reduce certain roles. The dataset I worked with was obtained from Kaggle and includes information such as job titles, industries, salaries, automation risk, projected job openings, and several demographic variables.

My analysis combines different approaches, including Exploratory Data Analysis (EDA), supervised learning models (Decision Tree, Random Forest, and Logistic Regression), and unsupervised learning through K-Means clustering. As I worked through these methods, I began to realize that not every dataset is suitable for accurate prediction, especially when the underlying features lack real-world variance. A major part of this project became understanding *why* machine learning models struggle with this synthetic dataset and what this means for using AI to forecast future job trends.

My main goal was to see whether it is possible to predict if a job is likely to increase or decrease in the future using the available features. Although I initially expected AI-related factors or automation risk to have some predictive power, the results turned out differently. This project helped me recognize not only the strengths and limitations of various machine learning techniques, but also how much the quality and structure of a dataset shape what we are able to discover.

2. Research Questions and Motivation

I was curious whether machine learning models could help identify which jobs might grow or decline between 2024 and 2030, and whether features such as salary, automation risk, education, or AI impact level play a role in these trends. At the same time, I wanted to practice a full end to end workflow: data cleaning, EDA, feature engineering, supervised learning, experiments, and clustering.

This project is guided by the following research questions:

- Which types of jobs appear to increase or decrease between 2024 and 2030 in this dataset?
- Are job trends related to features such as automation risk, AI impact level, education, or salary?
- Can we predict whether a job will increase or decrease using standard machine learning models?
- Why do these models struggle to achieve accurate results on this dataset?

- Do natural clusters exist within the data, and do they align with the Increase/Decrease labels?
- What can I learn from this analysis about how AI may affect IT and data-focused careers, including my own?

The goal of this project is not only to build predictive models, but also to better understand the limitations of synthetic forecasts and to reflect on how workers, including myself, can adapt their skills in an AI-augmented labor market.

3. Data and Basic Structure

The dataset I used in this project is called *"A Data-Driven Analysis of AI's Influence on Job Growth and Decline (2024–2030)"* and is available on Kaggle. It contains around 30,000 job records and includes a variety of features related to job roles, skill requirements, and projected employment trends. The below figure shows the main variables in the dataset:

```
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 13 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Job Title                    30000 non-null  object
 1   Industry                     30000 non-null  object
 2   Job Status                   30000 non-null  object
 3   AI Impact Level              30000 non-null  object
 4   Median Salary (USD)          30000 non-null  float64
 5   Required Education           30000 non-null  object
 6   Experience Required (Years)  30000 non-null  int64
 7   Job Openings (2024)          30000 non-null  int64
 8   Projected Openings (2030)    30000 non-null  int64
 9   Remote Work Ratio (%)        30000 non-null  float64
 10  Automation Risk (%)          30000 non-null  float64
 11  Location                     30000 non-null  object
 12  Gender Diversity (%)         30000 non-null  float64
dtypes: float64(4), int64(3), object(6)
memory usage: 3.0+ MB
```

Figure 1: Dataset structure

I also engineered an additional feature called **Percent_Change**, which measures relative growth between 2024 and 2030 job openings. This variable initially seemed useful for understanding future job trends, but later on I discovered that it contained extreme synthetic outliers.

When I first started this project, I applied several machine learning models right away, expecting that features like salary, automation risk, or AI impact level would help predict whether a job was likely to increase or decrease. However, when every model produced results close to random guessing, I realized that something was wrong, not with my models, but with the dataset itself.

After doing deeper exploratory analysis, I learned that many numeric features had **very narrow variance**. For example, most salaries were close to $90,000, automation risk was almost always around 50%, and even the projected job openings for 2030 showed unrealistic patterns. This homogeneity made it almost impossible for models to learn meaningful relationships.

Despite these limitations, working with this dataset was still valuable. It helped me understand how important data quality is, how to diagnose why models fail, and how to evaluate synthetic projections critically. Even though the models could not perform accurate predictions, the analytical process taught me lessons that I can apply in future real-world projects.

4. Methods

4.1 Data Preparation

Before building any models, I first prepared the dataset so that it could be used properly for analysis. This included cleaning the data, creating new variables, checking feature types, and transforming categorical columns.

I began by cleaning the dataset and creating a binary target variable where **1 = Increasing** and **0 = Decreasing**. I then applied one-hot encoding to categorical features such as industry, location, required education, and AI Impact Level. This allowed the models to use these fields in numerical form.

After encoding, I split the data into training and test sets using a 70/30 ratio. I only applied scaling to models that required it, such as Logistic Regression, while tree based models were trained on the unscaled data. This preparation step created a consistent and usable structure for all experiments that followed in the project.

	Median Salary (USD)	Experience Required (Years)	Job Openings (2024)	Projected Openings (2030)	Remote Work Ratio (%)	Automation Risk (%)	Gender Diversity (%)
count	30000.000000	30000.000000	30000.000000	30000.000000	30000.000000	30000.000000	30000.000000
mean	90119.965639	10.051433	5039.640833	5074.218100	49.836431	50.154229	49.979660
std	34412.013953	6.060678	2861.009654	2866.550722	28.966688	28.754889	17.274665
min	30001.860000	0.000000	100.000000	100.000000	0.000000	0.000000	20.000000
25%	60500.702500	5.000000	2570.000000	2586.750000	24.570000	25.400000	35.070000
50%	90274.115000	10.000000	5034.000000	5106.500000	49.570000	50.020000	49.885000
75%	119454.710000	15.000000	7527.000000	7573.000000	75.100000	75.030000	64.910000
max	149998.500000	20.000000	10000.000000	10000.000000	100.000000	99.990000	80.000000

Figure 2: The above table describing the dataset.

4.2 Exploratory Data Analysis (EDA)

During my exploratory data analysis, I reviewed the distribution of Job Status, summary statistics, and basic visualizations for all numeric features. The plots quickly showed that most variables, including salary, automation risk, experience, remote-work ratio, and gender diversity

had very limited variation across the dataset. I also noticed that the Percent_Change feature contained extreme values that looked unrealistic, suggesting that the underlying projections might be synthetic.

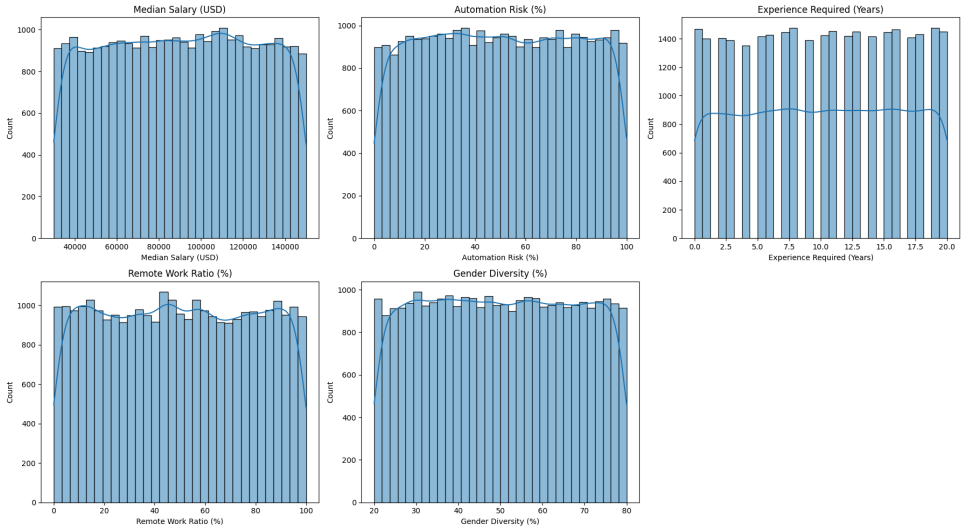


Figure 3: Bar charts : The bar charts show that the numeric features are spread almost evenly across their ranges. This indicates that the dataset does not reflect natural real-world distributions. Instead, the values appear artificially uniform, which makes it harder for models to learn meaningful patterns.

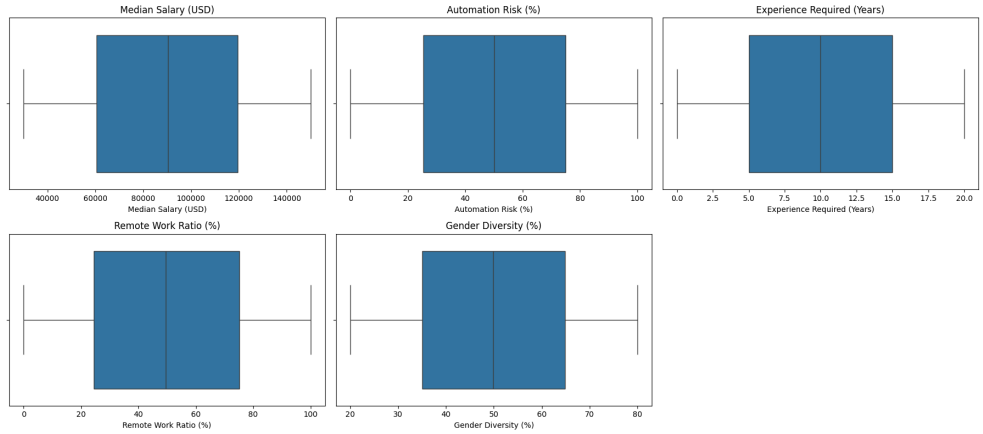


Figure 4: Boxplots: The boxplots highlight the same pattern: most numeric variables have very tight interquartile ranges and symmetrical shapes, reinforcing how uniform the dataset is. Percent_Change is the only variable with dramatic outliers, even after log-scaling.

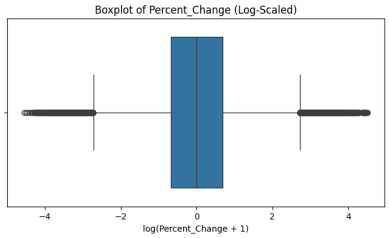


Figure 5: Boxplot of Percent_Change (Log-Scaled)

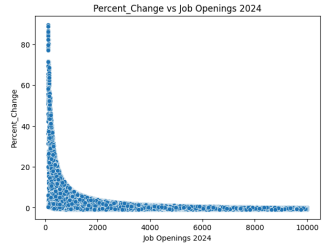


Figure 6: Percent_Change vs Job Openings 2024

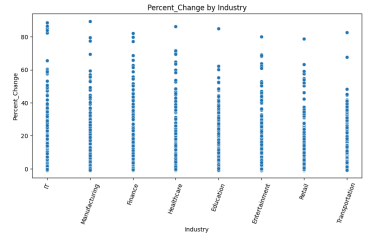


Figure 7: Percent_Change by Industry

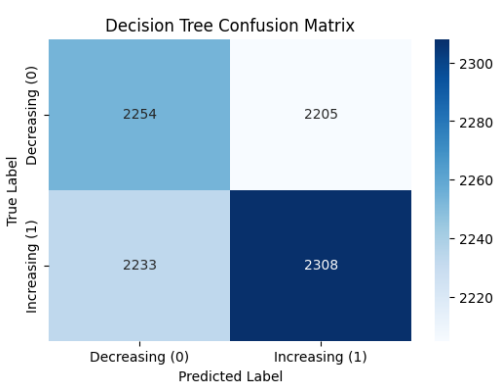
Overall, these plots suggest that the dataset lacks real variance and contains synthetic projections. This explains why the machine learning models later struggled to distinguish between Increasing and Decreasing jobs, and why Percent_Change introduced significant noise into the analysis.

4.3 Modeling Approach

To evaluate whether the dataset contained any predictive signal, I started by training three baseline supervised learning models:

- Decision Tree
- Random Forest
- Logistic Regression

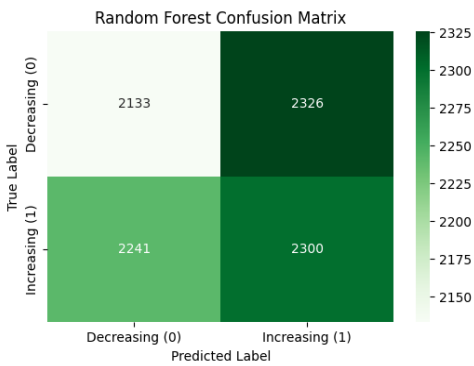
Decision Tree: The baseline Decision Tree model achieved around **0.50 accuracy** and a **ROC–AUC of 0.50**, which is essentially the same as random guessing. The confusion matrix below shows that the model is almost perfectly symmetric in its predictions: it correctly classifies roughly half of the *Decreasing (0)* jobs and half of the *Increasing (1)* jobs, while misclassifying the rest. This pattern indicates that the Decision Tree was unable to find meaningful splitting rules that separate the two classes. This result is consistent with earlier EDA findings, where features such as salary, automation risk, experience, and remote-work ratio showed almost no difference between Increasing and Decreasing jobs.




```
Decision Tree Accuracy: 0.5068888888888889

Classification Report:
              precision    recall  f1-score   support

           0       0.50      0.51      0.50      4459
           1       0.51      0.51      0.51      4541

    accuracy                           0.51      9000
   macro avg       0.51      0.51      0.51      9000
weighted avg       0.51      0.51      0.51      9000

ROC-AUC: 0.506876299212789
```

Figure 8: The figures above show the Decision Tree model results, including the classification report and the confusion matrix

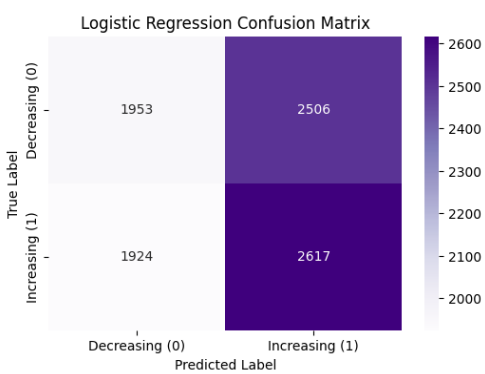
Random Forest Classifier: The Random Forest model performs almost identically to the Decision Tree, with an accuracy of about 0.49 and a ROC–AUC near 0.50, meaning it predicts no better than random guessing. Even with many trees and random feature selection, the ensemble is unable to separate Increasing vs. Decreasing jobs. The confusion matrix shows that for both classes, the model correctly classifies roughly half of the jobs and misclassifies the other half. This balanced but low performance indicates that the dataset lacks the predictive signal needed for any model to learn meaningful patterns, confirming that model complexity does not overcome the limitations of the underlying data.




```
Random Forest Accuracy: 0.4925555555555556

Classification Report:
              precision    recall  f1-score   support

           0       0.49      0.48      0.48      4459
           1       0.50      0.51      0.50      4541

    accuracy                           0.49      9000
   macro avg       0.49      0.49      0.49      9000
weighted avg       0.49      0.49      0.49      9000
```

Figure 9: The figures above show the Random Forest Classifier model results, including the classification report and the confusion matrix

Logistic Regression: The Logistic Regression model achieves about 0.51 accuracy and a ROC–AUC of 0.50, meaning it performs no better than random guessing. Even after scaling the numeric features, the model cannot find any meaningful linear relationships that separate Increasing vs. Decreasing jobs. The confusion matrix shows the same pattern as the other models: it correctly identifies only about half of the jobs in each class while misclassifying the rest. This confirms that there is no linear separability in the dataset and that the features do not contain the predictive signal needed for a linear classifier to learn useful patterns.




```
Logistic Regression Accuracy: 0.5077777777777778

Classification Report:
              precision    recall  f1-score   support

           0       0.50      0.44      0.47      4459
           1       0.51      0.58      0.54      4541

    accuracy                           0.51      9000
   macro avg       0.51      0.51      0.51      9000
weighted avg       0.51      0.51      0.51      9000

ROC-AUC: 0.5052068766794913
```

Figure 9: The figures above show the Logistic Regression model results, including the classification report and the confusion matrix

4.4 Experiment 1: Removing Percent_Change

From the baseline models, all accuracy and ROC–AUC scores were close to **0.50**, which suggested that the dataset might not contain meaningful predictive patterns. One strong suspicion from the EDA results was the **Percent_Change** feature, which showed extreme synthetic outliers ranging from hundreds to more than 10,000%. Such unrealistic values can distort tree splits and overwhelm model behavior. Because of this, I wanted to test whether removing this noisy feature would improve model performance.

In this experiment, I rebuilt the feature matrix **without Percent_Change** and retrained all three supervised models: Decision Tree, Random Forest, and Logistic Regression. Everything else remained the same, including the train–test split and preprocessing steps. This allowed me to isolate the effect of removing the outlier heavy feature.

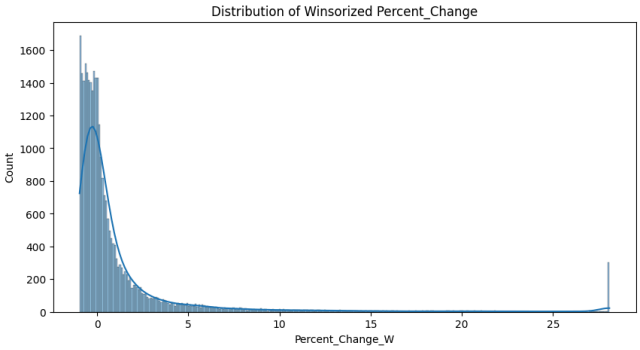
Removing Percent_Change did **not** improve performance. All models still produced accuracy and ROC–AUC values around **0.50**, nearly identical to the baseline. This outcome confirms that the problem is not caused by this feature alone, rather, the dataset itself lacks the predictive signal needed for any supervised model to learn meaningful patterns. The results support my earlier observations that most numeric variables have narrow variance and that Job Status is not strongly related to any of the provided predictors.

4.5 Experiment 2: Winsorizing Percent_Change

Although removing *Percent_Change* did not improve the baseline models, the feature still represents job growth information that could be useful if its extreme synthetic outliers are controlled. The raw values ranged from modest changes to more than **10,000%**, which distorted model behavior and dominated the overall variance. Winsorizing at the **1st and 99th percentiles** provided a way to smooth these unrealistic values while keeping most of the features intact.

In this experiment, I applied winsorization to cap the extreme ends of *Percent_Change* and included the winsorized feature (*Percent_Change_W*) in the model. I then retrained all three supervised models; Decision Tree, Random Forest, and Logistic Regression, using exactly the same pipeline and data split as before. This allowed me to isolate the effect of reducing outlier magnitude without removing the feature completely.

As a result, Winsorizing *Percent_Change* did **not** improve model performance. All models continued to show accuracy and ROC–AUC scores near **0.50**, identical to the baseline and Experiment 1 results. This indicates that the underlying issue is not the size of the outliers but the overall lack of predictive signal in the dataset. Even after trimming the outliers, the projections still behave like synthetic noise, and the features do not meaningfully distinguish between Increasing and Decreasing jobs.




```
Decision Tree (Winsorized) Accuracy: 0.5093333333333333
ROC-AUC: 0.5093209465931469
```

```
Random Forest (Winsorized) Accuracy: 0.49922222222222223
ROC-AUC: 0.4983181319891296
```

```
Logistic Regression (Winsorized) Accuracy: 0.5071111111111111
ROC-AUC: 0.5058867355853096
```

Figure 10: Distribution of the winsorized Percent_Change feature, showing smoothed extremes but the same long-tail synthetic pattern.

4.6 Experiment 3: K-Means Clustering

Since all supervised models performed at chance level, I used K-Means clustering to explore whether any natural structure existed in the dataset that the models could not capture. Unsupervised clustering allows us to look for patterns without relying on the "Increasing/Decreasing" labels, which appeared noisy and uninformative. If meaningful groups existed in the feature space, clustering would help reveal them.

To cluster initially, I standardized all numeric features and applied the Elbow Method to determine a reasonable number of clusters. The curve suggested a gradual decline without a clear elbow, but **k = 3** provided a practical balance between interpretability and detail. After fitting K-Means with three clusters, I visualized the results using PCA (two dimensions), examined cluster profiles, and compared cluster assignments to Job Status through a crosstab.

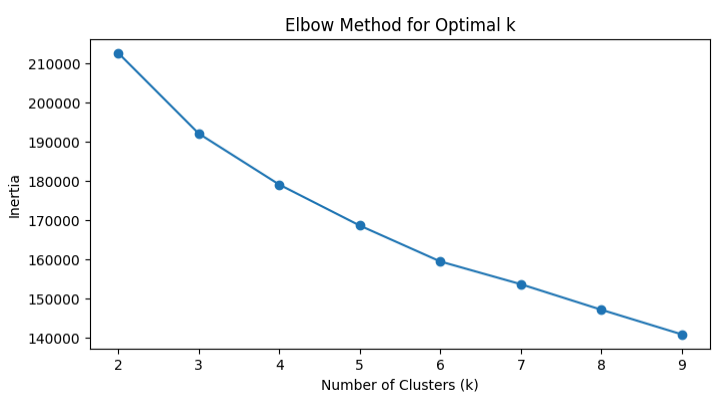


Figure 11 — Elbow Method : The Elbow Method shows a smooth, continuous decline in inertia without a sharp elbow, indicating weak cluster structure. A 3-cluster solution was chosen for interpretability

The clusters displayed only mild separation. The PCA projection showed that Cluster 1 (orange) shifts to the right on PC1, while Clusters 0 and 2 overlap heavily, indicating that most features contribute very little to differentiation. Cluster differences were subtle and driven mainly by one numeric dimension likely Percent_Change or projected openings. When compared to Job Status, each cluster contained almost exactly 50% Increasing and 50% Decreasing jobs, confirming that clustering did **not** uncover meaningful job-growth patterns. Overall, K-Means revealed minor numeric variation but no strong or interpretable structure related to job trends.

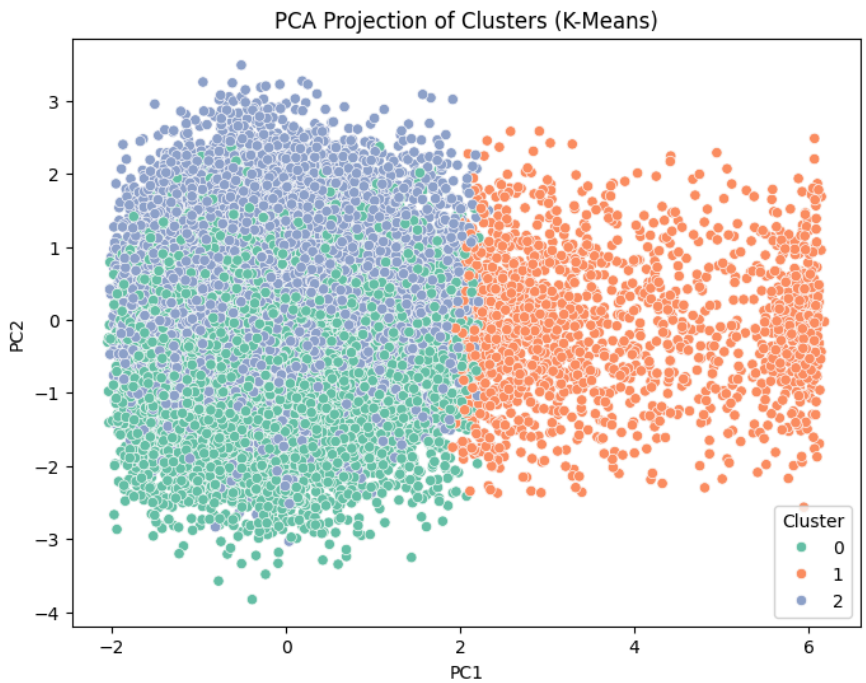


Figure 12 — PCA Projection of K-Means Clusters: The PCA scatterplot shows three clusters with only mild separation. Clusters overlap heavily, confirming that the dataset is highly homogeneous and does not contain strong natural groupings.

5. Results

Across all supervised learning models (Decision Tree, Random Forest, and Logistic Regression) accuracy and ROC–AUC scores remained close to **0.50**, which is equivalent to random guessing. None of the models were able to identify meaningful patterns that distinguish Increasing from Decreasing jobs.

Experiment 1 (Removing Percent_Change): Removing the Percent_Change feature did not improve performance. All scores stayed at approximately **0.50**, indicating that the extreme outliers in this feature were not the primary cause of poor results.

Experiment 2 (Winsorizing Percent_Change): Winsorizing the extreme projections also resulted in no performance gains. Even after reducing outlier magnitude, all models continued to behave like random classifiers.

Experiment 3 (K-Means Clustering): K-Means identified only weak cluster structure driven by small numeric differences such as salary and remote-work ratio. When compared with Job

Status, each cluster contained nearly **50% Increasing** and **50% Decreasing** jobs. This provided the strongest evidence that the Job Status label is **not explained** by the available features.

Overall, the results show that the dataset lacks predictive signal, and none of the supervised or unsupervised methods were able to uncover meaningful job-growth patterns.

6. Discussion and Limitations

The main finding of this project is that the dataset does not contain meaningful predictive signal for determining whether a job will increase or decrease in the future. Although it appears detailed at first, several issues limit its usefulness for modeling.

First, the target variable (*Job Status*) behaves like a synthetic label. Key features such as salary, automation risk, experience, gender diversity, and remote-work ratio show almost identical distributions for both Increasing and Decreasing jobs, leaving no natural separation for models to learn from.

Second, most numeric variables have extremely narrow variance. Salaries cluster around $90k, automation risk around 50%, and experience around 10 years. This uniformity prevents both linear and tree-based models from identifying meaningful patterns.

Third, the 2030 job projections appear unrealistic. The *Percent_Change* feature includes extreme synthetic growth values, sometimes thousands of percent, which introduce noise and instability. Even after winsorizing or removing this feature, predictive performance remained unchanged.

Finally, unsupervised clustering confirms the same pattern: each K-Means cluster contains almost exactly **50% Increasing** and **50% Decreasing** jobs, indicating that the dataset's internal structure does not align with the target label.

Overall, the limitations come from the dataset itself rather than from the modeling process. This project reinforced an important lesson: **machine learning cannot extract signals where none exists**, and high-quality data is essential for meaningful prediction.

7. Conclusion and Future Work

This project explored how AI might affect future employment trends using a synthetic job-market dataset. Although the dataset included many features, none of them meaningfully predicted whether a job would increase or decrease. All supervised models performed at around **50% accuracy**, even after removing or winsorizing Percent_Change, and clustering showed no relationship between natural groups and the Job Status label. These results show that the dataset lacks real predictive signal.

Despite this, the project was useful for practicing a full data-science workflow and understanding how to diagnose modeling failures. It reinforced an important lesson: machine learning cannot produce meaningful predictions when the underlying data does not reflect real-world patterns.

Future work could include using real labor-market data (such as BLS projections), creating rule-based labels based on economic theory, analyzing job postings with NLP, or focusing on specific fields like IT or data science where AI-related skill trends are clearer.

8. References

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